

familiar: irrelevant results, too few or too many results. This happens because a search engine does not know exactly what you are searching for. Think of a search engine as an agent that you send on a mission. You need to communicate with it for it to do its job properly, but you can only communicate using a few choice words. This is the problem. As of now, we cannot tell a search engine to 'find me all pages that talk about data acquisition techniques being pursued after 2002'. We have to use the keywords: 'data acquisition techniques'. The problem is two-fold: first, contextualisation—the search engine needs to understand what you're searching for; and second, insufficient tagging—Web pages need to be tagged with metadata, so that a search engine knows that a page contains what you're looking for.

Regarding contextualisation, we might envisage a search engine in the future that will do natural language processing (NLP), and figure out what you mean, when you enter a query in English. This is one approach that

Voice Search

Search, for most of us, means only text-input based search, but it is now possible to use audio to search. The best application of this would be to find music in a collection: often, we find ourselves in a situation where we can hum the tune of a song but can't recollect its name. Progress is being made towards a situation where you hum the tune into a mic, and the system reports the closest matches. For example, Philips has developed a technology called Audio Fingerprinting. The company claims that the technology can be used to give complete information on a song such as name, artist, and album, by sending three seconds of a song to its server.

A cell phone user would be able to hold the mobile's receiver near the music source, and get the server to respond with information on the song. Google is also in the process of developing voice-based search: in its current form, the user needs to dial a number, speak out keywords, and then click a link at <http://labs1.google.com/gvs.html>. The results are displayed in a new window.

How Search Engines Work

Crawlers, or spiders, are the programs that most search engines use to document the Web. Some search engines do not use crawlers, relying instead upon human-fed information about pages, while others use a combination of both. Crawlers visit Web sites to read the information in them—both the content as well as the meta-content, or meta-tags, and follow the links that the site points to. The information that the crawler gathers is sent back to a central database where it is indexed; this indexing is what enables the pages to be retrieved upon demand, when you do a search. Crawling is a continuous process—crawlers work all the time, 24 hours a day, 7 days a week, visiting and re-visiting pages so that the indexed information is up-to-date.

Given keywords, a search engine scans its collection of indexes to see if the keywords are present. If they are, the engine uses metadata to determine how relevant each document that contains those strings is likely to be. How search engines differ is in the algorithm they use for their crawlers. Google uses an algorithm called PageRank to map the Web. PageRank checks to see how many pages link to a given page, and increases the 'importance' of a page if more pages link to it. This page is more likely to show up in search results. Also, the more 'important' the pages that link to this page are, the more important this page becomes, and so on. Thus, the most important page is the one that is linked to by several other important pages.



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I don't personally see any major breakthrough in the next five years, but I see existing technologies becoming more widespread and people becoming more dependent on them

string refers to. When you launch a search for 'Introduction to Java', the results would also contain pages described, or tagged as, 'Java tutorials'.

Alternative Search

Latent Semantic Indexing (LSI) searches by looking at the meaning of the

some researchers find an interesting possibility, but solving the NLP problem is hard. NLP is being put to use to an extent, though, in IBM's WebFountain programme.

Consider when you need information on 'Kolkata'. You'd probably enter the search string 'Kolkata information', or something similar. The results will not contain pages that refer to the city as 'Calcutta' without an instance of the word 'Kolkata', but you, on the other hand, would expect these pages to appear in the results. This is another limitation of current search engines—they only look for exact word-to-word matches, and don't understand the meaning of your search string. This is where semantic search comes in. Semantic searching allows you to retrieve information that is similar in meaning to what the entered search

words, not just by matching keywords. The first step is to extract meaningful and relevant words from a document—for example, in the sentence 'The car was developed by the famous Indian design house', the important words would be 'car,' 'developed,' 'Indian,' and 'design'. Similarly, keywords need to be extracted from the document. Once the keywords are brought together, the Term-Document Matrix (TDM) is generated. In this current form, the TDM only represents the presence or absence of a keyword in a document, but linguistically some keywords will have more importance than others. This is performed in the next step, called Term Weighting. The 1s in the matrix are modified to reflect the importance of the keyword in the document. Consider the keyword 'DDR RAM', appearing in two documents, one explaining DDR RAM, and the other listing new RAM